

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591 -> 2.365762079	delta=+0.0141645
ratio=1.0060233		
CoG x [m]	-0.002024708562 -> -0.002188239649	delta=-0.000163531
ratio=-		
CoG y [m]	-3.052657894e-05 -> 0.0008604997098	delta=+0.000891026
ratio=-		
CoG z [m]	0.009509749599 -> 0.01054183284	delta=+0.00103208
ratio=-		
rotor effectiveness 1	1 -> 0.9669407746	delta=-0.0330592 ratio=0.96694077
rotor effectiveness 2	1 -> 0.9719416876	delta=-0.0280583 ratio=0.97194169
rotor effectiveness 3	1 -> 0.9574778107	delta=-0.0425222 ratio=0.95747781
rotor effectiveness 4	1 -> 0.9560050918	delta=-0.0439949 ratio=0.95600509
rotor command lag [s]	0.004997730255 -> 0.2	delta=+0.195002 ratio=-

Inertia matrix [kg m^2]

reference:

[0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[1.901508003e-05, 5.9167305e-08, 0.1289921107]

estimated:

[0.0650101068, -2.371528734e-05, 2.300286326e-05]
[-2.371528734e-05, 0.06524486925, 5.49647582e-06]
[2.300286326e-05, 5.49647582e-06, 0.1289937364]

Principal moments: reference -> estimated

principal moment 1 [kg m^2]	0.06495264516 -> 0.0650077265	delta=+5.50813e-05
ratio=1.000848		
principal moment 2 [kg m^2]	0.06500006701 -> 0.0652472408	delta=+0.000247174
ratio=1.0038027		
principal moment 3 [kg m^2]	0.1289921163 -> 0.1289937451	delta=+1.6288e-06
ratio=1.0000126		

Optimization

joint dynamics loss: 1.1006756753
Gaussian-prior cost: 0.332470685209
selected strict-ZOH rotor lag [s]: 0.2

Bag failure_1_single

reconstruction position RMSE [m]: 0.000497212640178
reconstruction position component RMSE [m]: [0.000302 0.000297 0.000261]
reconstruction orientation RMSE [deg]: 0.528821719301
gyro consistency RMSE xyz [rad/s]: [0.060797 0.087546 0.042578]
specific-force consistency RMSE xyz [m/s^2]: [0.936944 0.20288 0.378619]
inferred external force RMS xyz [N]: [0.969383 0.533609 3.016821]
inferred external torque RMS xyz [N m]: [0.121804 0.283217 0.21073]
bag dynamics loss: 1.1006756753
gimbal command lag [s] 0.004998683929 -> 0.03866942712

Recorded command timestamp intervals (data-derived)

Bag failure_1_single
rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s
gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s

Selected optimizer diagnostics

status=1 success=True nfev=4 njev=4
message=`gtol` termination condition is satisfied.
reported optimality=5.663680902046053e-08
initial cost=1.4331464615421048
final cost=1.4331463605116408
cost reduction=1.0103046399478899e-07
coordinate step L2=0.0002695517511298434
final gradient inf-norm=5.663680902046053e-08
gradient tolerance satisfied=True
final Jacobian rank=14
final Jacobian cond(nonzero)=408.250079270499
max directional finite-difference Jacobian relative error=9.312112664126403e-12